

Figure 3: Gluster Storage cost effectiveness (Source: <https://redhatstorage.redhat.com/2016/11/03/idc-the-economics-of-software-defined-storage/>)

RGHS has a great feature called ‘snapshotting’, which helps in taking ‘point-in-time’ copies of Red Hat Gluster Storage server volumes. This helps administrators in easily reverting back to previous states of data in case of any mishap.

Some of the benefits of the snapshot feature are:

- Allows file and volume restoration with a point-in-time copy of Red Hat Gluster Storage volume(s)
- Has little to no impact on the user or applications, regardless of the size of the volume when snapshots are taken
- Supports up to 256 snapshots per volume, providing flexibility in data backup to meet production environment recovery point objectives
- Creates a read-only volume that is a point-in-time copy of the original volume, which users can use to recover files
- Allows administrators to create scripts to take snapshots of a supported number of volumes in a scheduled fashion
- Provides a restore feature that helps the administrator return to any previous point-in-time copy
- Allows the instant creation of a clone or a writable snapshot, which is a space-efficient clone that shares the back-end logical volume manager (LVM) with the snapshot

BareOS configured on GlusterFS has the advantage of being able to take incremental backups. One can create a ‘glusterfind’ session to remember the time when it was last synched or when processing was completed. For example, your backup application (BareOS) can run every day and get incremental results at each run.

More details on the RGHS snapshot feature can be found at <https://www.redhat.com/cms/managed-files/st-gluster-storage-snapshot-technology-overview-inc0407879-201606-en.pdf>.

The best open source backup software tools

AMANDA open source backup software

Amanda or Advanced Maryland Automatic Network Disk Archive (<https://amanda.zmanda.com/>) is a

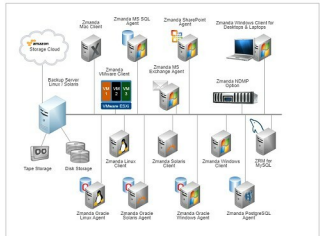


Figure 4: AMANDA architecture

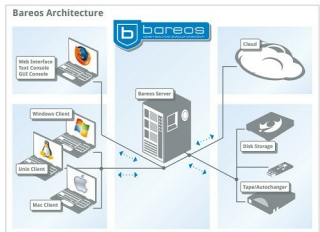


Figure 5: Bareos architecture

popular, enterprise grade open source backup and recovery software. According to the disclosure made by AMANDA, it runs on servers and desktop systems containing Linux, UNIX, BSD, Mac OS X and MS Windows.

AMANDA comes as both an enterprise edition and an open source edition (though the latter may need some customisation). The latest version of the AMANDA Enterprise version is release 3.3.5.

It is one of the key backup software tools to be implemented in government, databases, healthcare and cloud based organisations across the globe.

AMANDA has a number of good features to tackle the explosive data growth and for high data availability. It provides and helps in managing complex and expensive backup and recovery software products.

Some of its advantages and features are:

- Centralised management for heterogeneous environments (involving multiple OSs and platforms)
- Powerful protection with simple administration
- Wide platform and application support